

Point-Wise Contraction-Based Control Based on Vector-Space Optimization under State and input Constraints

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Abstract—

I. INTRODUCTION

- Advantages of contraction-based control
- Challenges in existing methods
 - Absence of explicit handling of constraints
 - Requirements of prior computation of contraction metrics
 - Limitations to online optimization in real-time applications (computational complexity)
- Potential of vector-space optimization
 - Integration of constraints directly into the optimization framework
 - Input constraints and state constraints can be handled explicitly
 - CBF based constraints can be easily integrated

II. PRELIMINARIES

III. MAIN RESULTS

IV. NUMERICAL VALIDATIONS

V. CONCLUSIONS

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